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Chapter 1 Brief introduction of quayside crane

Summary：

To a great extent, the lifespan of the crane and its auxiliary components is determined by **applying properly, operating safely and maintenance periodically**.

As the start chapter of this new edition of construction and maintenance manual for quayside container cranes, it aims at providing the fundamental knowledge about the construction, functions, performances and features of the crane for the operators and maintenance personnel in order that they can conduct the operating and maintenance in proper and safe way.

This chapter will be organized in a manner of figures, photographs and text to describe the Quayside Container Crane, including the general features, structure constructions and the compositions, layout, technical parameters, functions, characteristics of the mechanisms. The four main mechanisms, the Main Hoist, Trolley Drive, Boom Hoist and Gantry Travel, will be the primary part and detailed. Furthermore, the auxiliary components of the crane will be included as well.

1.1 Outline and main parameters for quayside container crane

Since the construction of all the Quayside Container Cranes are identical, all detailed information of the common mechanisms in this chapter will be based on an example Quayside Container Crane (hereinafter call the "Crane") which was designed and manufactured by ZPMC for KICT. This crane features of double-boxes boom & girder, rope propelled trolley and continuous catenaries trolley support system. It has to be emphasized in particular that all the parameters appeared in this chapter are unusual and possessed in this project. The main parameters are shown in below table:

Table 1-1: Main Parameters for Quayside Container Crane

Basic Parameters	Rated Load Capacity(t)	Under Spreader	65	Speed	Hoist Speeds（m/min）	Full Load	75
		Under Hook	75			No Load	180
	Lift Height (m)	Above Rail	45		Trolley Travel Speed（m/min）		244
		Total Lift Height	63		Gantry Speed（m/min）		46
	Outreach（m）	70		Accelerating Speeds	Main Hoist Acceleration Time（sec）	Full Load	2.5
	Back reach（m）	20				No Load	6
	Max Operating Wheel Load（t）	Sea-side	87		Trolley Travel Acceleration Time（sec）		5
		Land-side	57		Gantry Travel Acceleration Time（sec）		8
	Max Stowed Wheel Load（t）	Sea-side	75				
		Land-side	106				
	Inside Clearance Between Legs(m)		>18.3	Boom Hoist time One Way（min）≤	0°-80°(Stowed Position)		~5
	Height Under Link Beam(m)		12.5	Motor Power	Hoist Motors	Quantity（pcs）	2
	Overall Crane Width from Bumper to Bumper（m）		27			Power(Kw)	450
	Crane Weight（t）		~1450			Rotating Speed(rpm)	750/1800
	Setback of WS Trolley Girder Support Beam（m）		2		Trolley Motors	Quantity（pc）	1
Special Functions	Anti-Snag	Provided				Power(Kw)	275
	Type of Anti-Sway	Electrical				Rotating Speed(rpm)	1750
					Gantry Motors	Quantity（pc）	16
						Power(Kw)	20
						Rotating Speed(rpm)	1750
Spreader Parameters	Trim	±5°			Boom Hoist Motor	Quantity（pc）	1
	List	±3°				Power(Kw)	375
	Skew	±5°				Rotating Speed(rpm)	1750
	Single 20′/40′/45′ & Twin-20′ Telescopic Spreader			Gantry Main Data	Rail Gauge(m)		30
Trolley Main Data	Rail Guage(m)	6.3			Wheel Base (m)		15.8
	Wheel Base(m)	4.44			Wheel Diameter(mm)		Φ800
	Wheel Diameter(mm)	Φ630mm			Wheel Number on Each Leg(pcs)		8
	Wheel Number	4			Driving wheels on Each Leg（pcs）		4
	Weight(t)	24					
Stowage Pin Between Crane and Wharf		Manual		Double Boom Hinges Adopted between Boom and Trolley Girder			
Stowage Pin Between Boom and A-Frame		Automatic		Power Supply on Wharf	11KV,50HZ		

1.2 Configuration of the crane steel structure

With a huge volume profile and suffering severe impact and frequently start-up and braking, a crane is obliged to ensure container positioned accurately and handled efficiently at high speed during operation. All these are ensured by the well-designed steel structure and the advanced control system. Most of the structure member section is trunk or box except some in tubular shape. The members of Main structure refer to Figure 1-1.

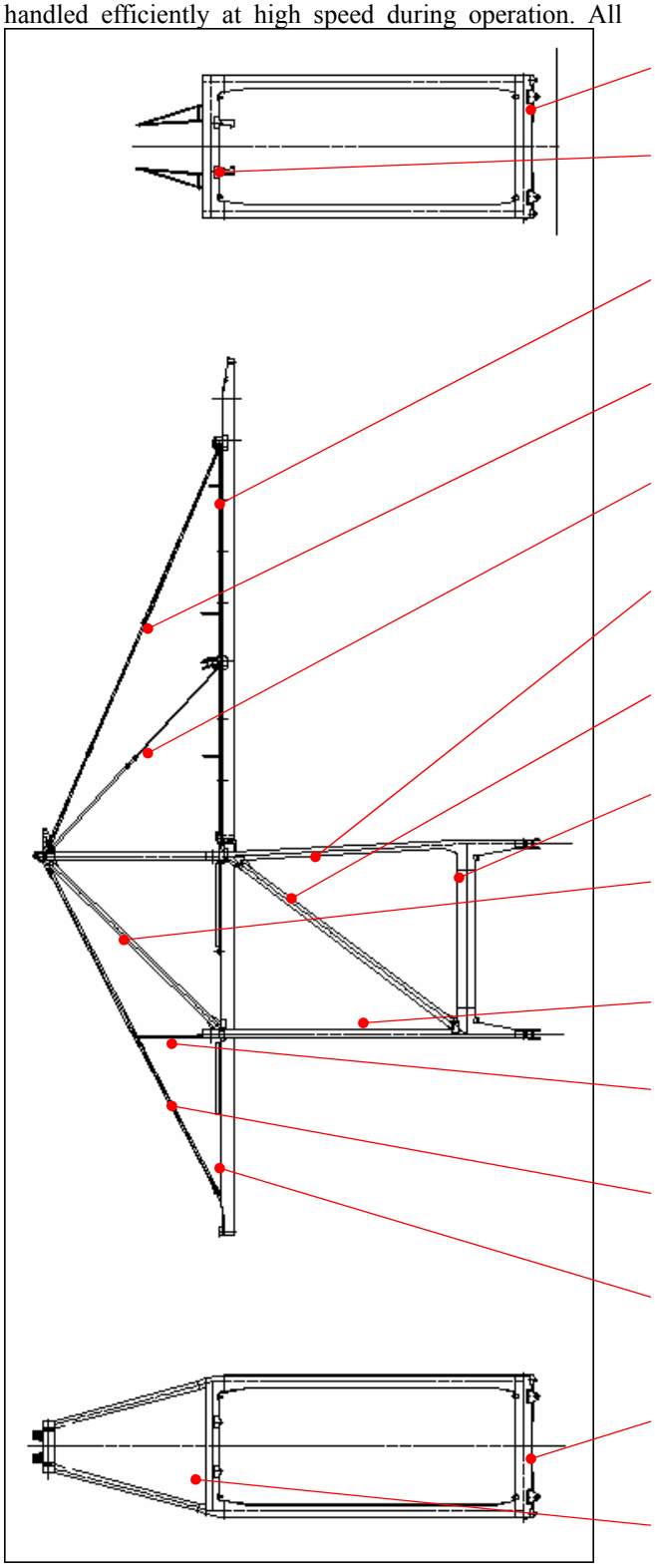


Figure 1-1: Crane Steel Structure Overview

1 Sea-side TGSB
2 Sea-side Sill Beam
3 Trolley Girder
4 Trolley Girder Backstay
5 Support of Trolley Girder Backstays
6 Land-side Legs
7 Backstays of Apex Beam
8 Portal Beam
9 Frame bracing
10 Sea-side Legs
11 Inner Boom Forestay
12 Outer Boom Forestay
13 Boom
14 Land-side TGSB
15 Land-side Sill Beam

1.3 Drive mechanism on crane

The four major drive mechanisms, Main Hoist Drive, Boom Hoist Drive, Trolley Travel Drive and Gantry Travel are provided on the crane and sheltered in a weatherproof Machienry House except the Gantry Travel as shown in the below Figure 1-2. Alternatively, the Trolley Travel Drive will be incorporated with the travelling trolley outdoors if it is self-propelled type.



Figure 1-2: Major Drive Mechanisms Layout in Machinery Room

1 boom hoist drive 2 trolley travel drive 3 main hoist drive

1.3.1 Main hoist drive

The main hoist drive is located in the machinery house. It is powered by two variable frequency AC motors connected to the reducer input shafts through the flexible elastomeric type couplings (high speed). Two rope drums are connected to the output shafts of the reducer through the two flexible gear couplings (lower speed). The driving force of this equipment is transmitted from the motors to the drums by the high speed couplings and drum couplings and amplified through the reducer driving the drum and rope to lift or lower the cargo attached under the telescopic spreader or hook beam. The data (Table 1-2) of main components refers to (Figure 1-2).

Table 1-2: Main Data of Major Components for Main Hoist Drive

Motor	Type	BDK-1KMU(YASKAWA)	Drum Brake	Type	SB315
	Power	2X450kw		Brake Torque	2X254.5kNm
	Rotating speed	750/1800rpm	Motor Coupling	Type	AFN SB160-900x80
Reducer	Type	ML3PSF130E-1600(SEW)		Rated Torque	19.2kNm
	Ratio	18.551		Brake Torque	38.4kNm
	Rated Output torque	2X212.6kNm	Drum Coupling	Type	DC08B
Motor Brake	Type	YP41-ED4500-80-900X30		Rated Torque	315kNm
	Braking Torque	2x23000Nm	Wire Rope	Diameter	Φ31.5mm

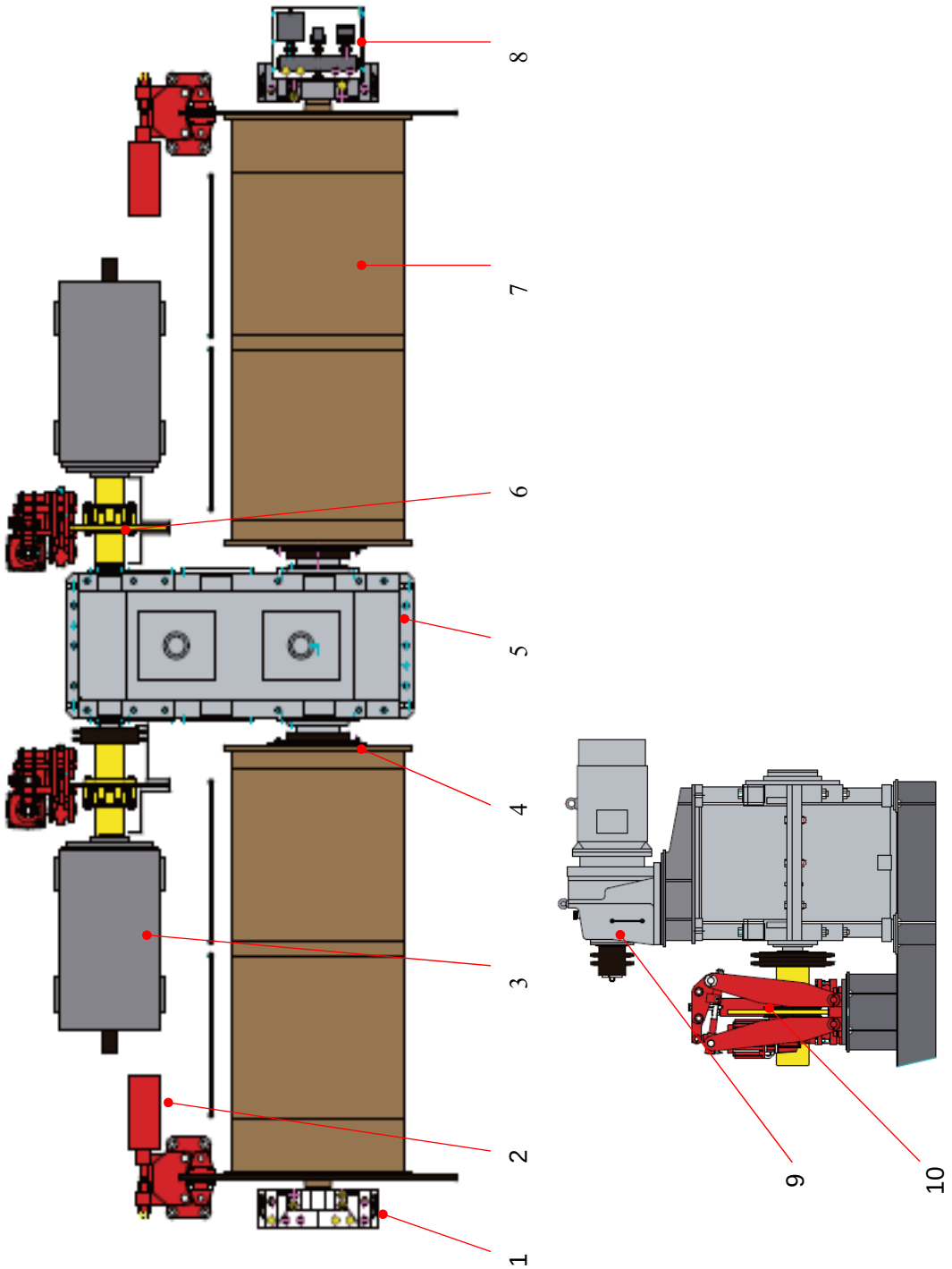


Figure 1-2: Scheme and Photograph for Main Hoist Drive

1 Drum Bearing Seat
2 Drum Brake
3 Motor
4 Drum Coupling
5 Reducer
6 Motor Coupling
7 Drum
8 Limit Switch Device
9 Emergency Device
10 Motor Brake

	Installed at the end of the main motor, an emergency drive is capable of hoisting or lowering the spreader slowly in case the hoist motor or drive control system failed and can not be fixed immediately. Upon the crane loss of high voltage power while hoisting or lowering, this emergency device can be plugged on the shore power and continue the operation till the container removed from the spreader or ready for next step.

1.3.1.1 Main hoist reeving system

The wire ropes reeled out from the drum, then run through the following route:

Drum→Trim/List/Skew/Snag System on girder rear end → Sheaves on trolley frame → Sheaves on head-block → back to Sheaves on trolley frame → Load cell at boom tip

The rope ends are tied with four load cells which are mounted under the sheave blocks respectively at the boom tip (refer to Figure 1-4). The signal will be generated by the tension force along the wire ropes then transmitted to PLC (Programmable Logic Controller) and CMS (Crane Management System) so that the hanging load and its eccentricity information can be figured out and recorded.



Figure 1-4: load cell location

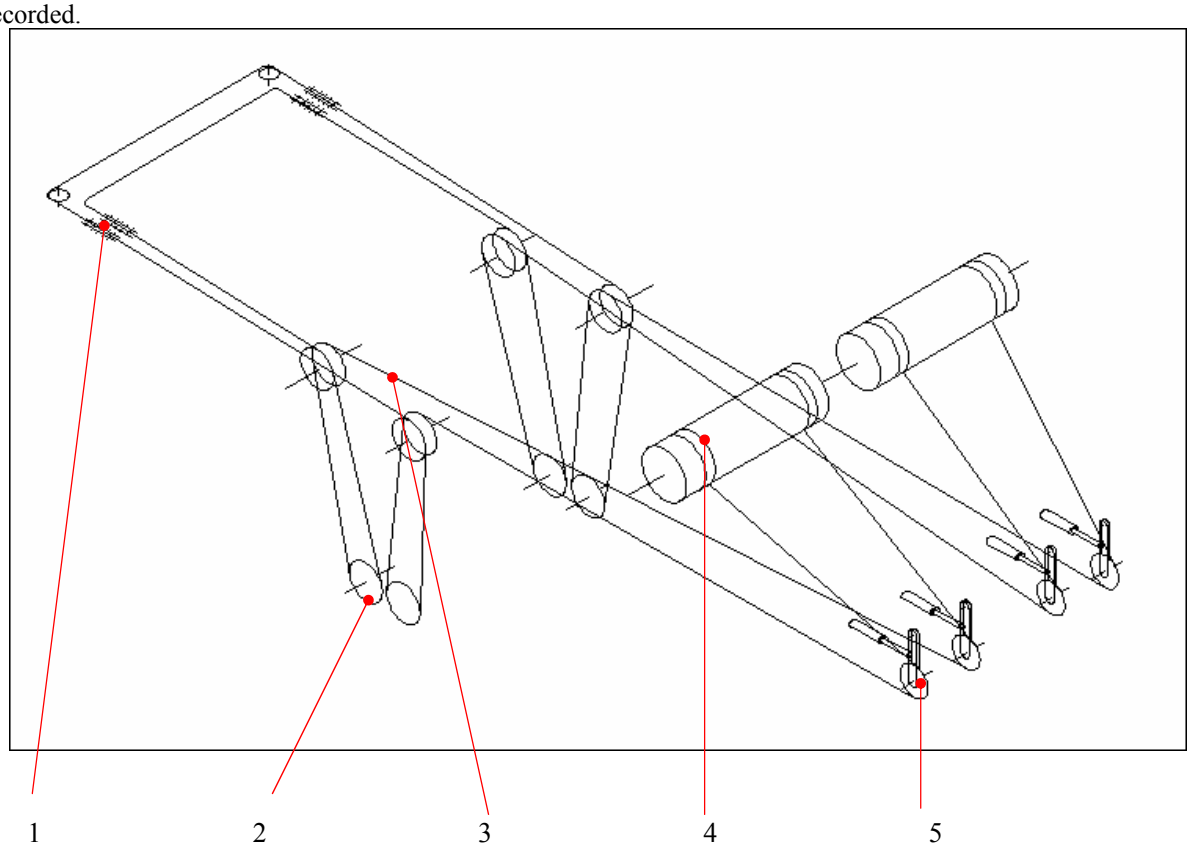


Figure 1-5: Main Hoist Rope Reeving System (Continuous Catenary Roller Support System)

1 Load cell on Boom Tip

2 Sheave Block on Head-block

3.Traveling Trolley

4 Drum of Main Hoist Drive

5 Trim/List/Skew & Anti-Snag on Girder Rear End

1.3.1.2 Interlocks for main hoist drive

1） Over-speed switch
External over-speed protection for the main hoist is provided at the end of one drum. Actuation of the switch initiates an emergency stop. The theoretical trip speed setting for the switch is 110% of the motor maximum speed. In order to resume the crane operation, the switch must be manually reset from its back once it is tripped. (Refer to Figure 1-6) As an optional device, which is connected with the motor shaft end, the incremental encoder can transmit the real-time speed signal then PLC will evaluate over-speed or not. (Refer to Figure 1-7)



Figure 1-6: Limit Switches at Drum End of Main Hoist Drive

1 Cam limit Switch 2 Encoder 3 Over-speed Switch

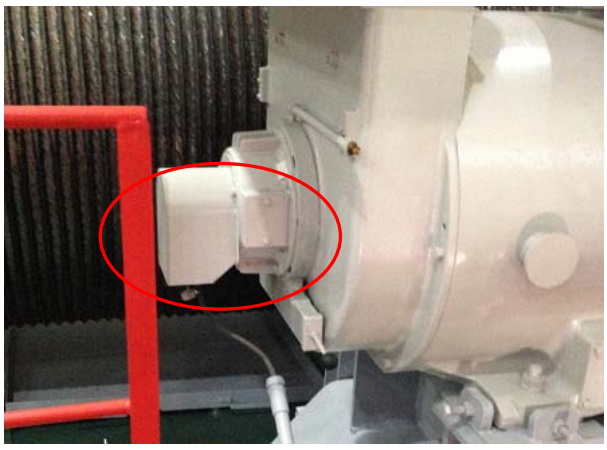


Figure 1-7: Incremental Encoder

2） Motor brake limit switch
Each of the two hoist motor brakes is monitored by a brake released limit switch (one normally open contact) and a brake manually-released limit switch (one normally-closed contact). Without signal of hoisting from joystick, the hoist motion is inhibited from operating as long as either switch is actuated.

3) Drum brake limit switch
Each of the two hoist drum brakes is monitored by a brake released limit switch (one normally open contact). Without hoisting signal from joystick, the hoist motion is inhibited from operating as long as either switch is actuated.

4) Motor thermal switch
Each of the two hoist motors features an embedded over-temperature switch (one normally closed contact). Actuation of the switch indicates the motor temperature is at fault level. The hoist motion is inhibited from operating as long as either of the switches is actuated.

5) Load weighing system
Information for the lifted load is provided by the weighing system. The hoist drives also provide overload indications. Refer to Section 1.3.1.1 for more detailed description

6) Anti-Snag system
When the snag occurs, the hydraulic system will enable the snagged cylinder to absorb the inertia energy of the mechanical moving parts through releasing its fluid and pressure. At the same time, an emergency stop will pause the hoist moving as long as the drive and PLC receives the signal sent from the electrical overload transducer or the pressure switches mounted on the snagged cylinders. The total snag event takes place in less than a second. Lowering the hoist is allowed at reduced speed after first resetting control power to release the strain on the ropes. No hoist raise motion is permitted until the event is cleared.

7) Boom interlocks
If the boom is not fully at horizontal position (stowed or maintenance), the hoist motion is disabled. When the boom is raised the hoist functions at the reduced speed (empty spreader only)

8) Spreader operation modes
The hoist and head-block / spreader system operates in three different modes, container, cargo and reeve.

9) Spreader mode interlocks (Slack rope protection)
In case all the proximity switches around the twist locks are activated (means cargo already landed), further lowering motion of the spreader will be inhibited. These switches contribute to preventing the hoisting ropes from slacking. Hoisting motion is permitted when all these switches are released or activated.

10) Spreader cable reel interlock (for motorized cable reel only)
Whether improper tension, irrational position or selection of manual mode at the trolley top control station, the main hoist operation will be inhibited. The cable reel runs in two modes, auto and manual. Auto should be selected if operating from the cab

and manual selected if from the local control station settled on the trolley. Hoist movement, raise and lower will be disable if the reel manual mode is selected in both container mode and cargo mode.

1.3.1.3 Positioning Limits for Main Hoist Drive

Normal hoist limits are controlled by the absolute encoder mounted on the main hoist motor. The home (or zero) position of the hoist is located at the upper limit of normal travel, while the hoist height indication is still counted from wharf ground level.

In addition to the absolute encoder limits, there is another weight operated lever switch (as shown in Figure 1-8) mounted on the bottom of the trolley. Such redundant design provides the over travel protection in a better and safer way.



Figure 1-8: Rod under Trolley Frame for the Weight Operated Lever Switch

1） Hoist over travel stop
The weight operated lever switch work together with the cam switch to control the hoist over travel stop. Either actuation of these two limit switches will initiate an emergency stop to the main hoist motors, and will reset the absolute encoder (zero). The “limit back-out bypass” selector switch in the electrical house permits only lowering down.

2） Up stop verification (cam limit switch)
The up limit is controlled by the incremental encoder together with the cam switch. An emergency stop will be initiated as long as the hoist speed signal transmitted by the incremental encoder is not zero.

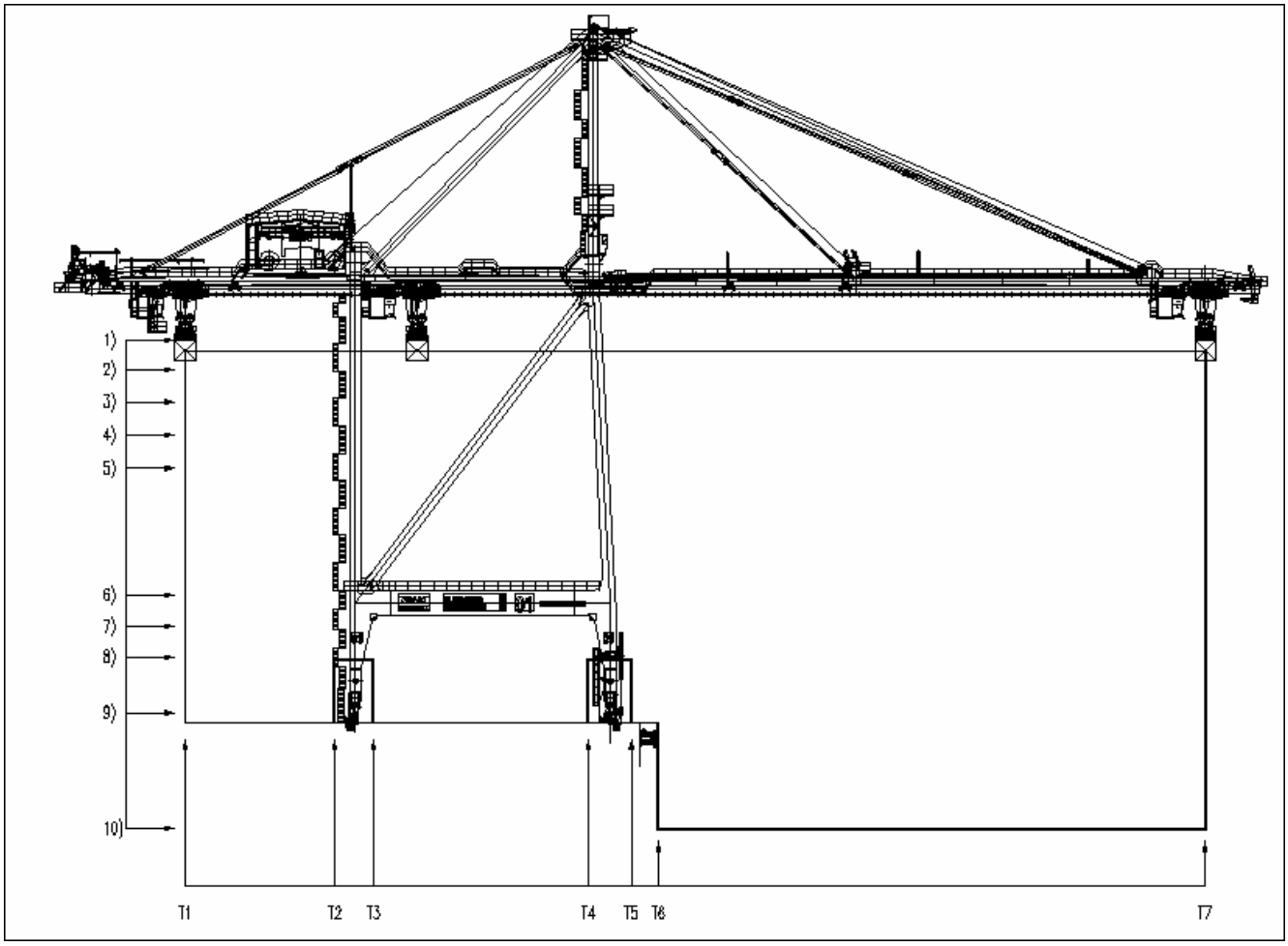
3） Raise normal travel stop
Controlled by the incremental encoder together with the cam switch, a ramp stop of the hoist motors will be initiated by either switch actuation and the absolute encoder can be reset manually from the CMS.

4） Up slowdown verification
Controlled by the incremental encoder together with the cam switch, an emergency stop will be initiated if the up slowdown contact of the cam switch being activated meanwhile the up hoisting speed reflected by the incremental encoder still not slowdown to the preset range.

5） Lower slowdown verification (over dock)
This is similar to the Up Slowdown Verification.

6） Lower normal end stop
This is similar to the Raise Normal Travel Stop.

7） Lower over travel verification
The absolute encoder and cam switch simultaneously control and protect the operation of lowering down in the cell.



T1 T2 T3 T4 T5 T6 T7

Figure 1-9: Scheme of the Functions for Hoist Traveling Limit

1.3.1.4 Trim-list-skew and snag protection device

These functions are accomplished by the multi-functional hydraulic system at girder end tie, which is composed of four hydraulic cylinders and valves power-pack. The hydraulic cylinders are attached to the four hoist ropes by turning sheaves.

The hoist rope provides spreader positioning (trim/list/skew) and snag load protection through the multi-functional hydraulic system.

Spreader tilting is accomplished by manipulating the position of the cylinders, lengthening or shortening the rope falls to adjust the container tipping angle around its three orthogonal axes.

As snag event occurs, the kinetic energy of the hoist mechanism that is rotating at high speed is transformed into heat energy and releases it by the hydraulic system, which prevents the disaster such as wire rope broken, crane structure deforming or even whole crane overturning or collapsing.

The cylinder position is monitored by linear displacement transducers located inside each of the cylinder rods. Output from each of the cylinders is directly proportional to the 1400mm stroke of the cylinders. The initial home position, the minimum extruding under working condition, is at least 375mm for each of the cylinders. That is the stroke 0mm ~ 650mm for the snag protection, and stroke 650mm ~ 1400mm for the spreader Trim-List-Skew motions and/or compensating the length variety for the hoist wire ropes.



Figure 1-30 Spreader Trim-List-Skew and Snag Protection Device on the Girder Tie End

1） Trim, list and skew motions

By actuating combinations of hydraulic cylinders motions in different directions, this device adjusts the TRIM, the LIST and the SKEW.

- TRIM

Tipping or rotating of the spreader around a horizontal axis perpendicular to the gantry rail. From the operator's perspective, trim raises and lowers the left or right end of the spreader as shown in Figure 1-11.

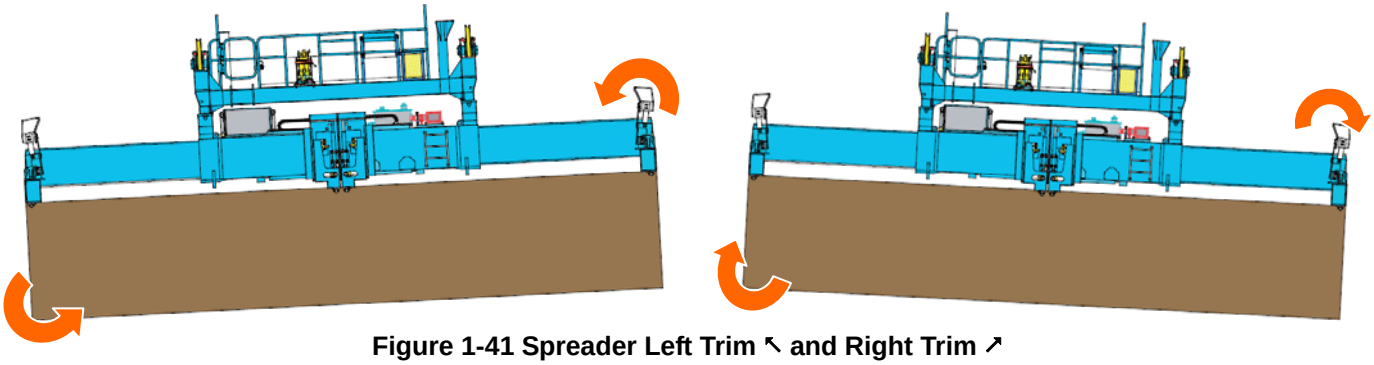


Figure 1-41 Spreader Left Trim ↖ and Right Trim ↗

- LIST Tipping or rotation of the spreader around an axis parallel to the gantry rails.



Figure 1-52 Spreader Forward List ↖ and Backward List ↗

- SKEW Rotation of the spreader around a vertical axis.

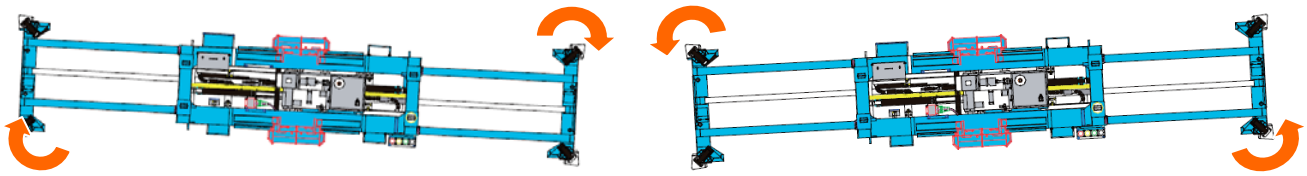


Figure 1-63 Scheme of Spreader Clockwise Skew ↖ and Counter Clockwise Skew ↗

When the Trim/List/Skew is at their maximum angles, the cylinders cannot actuate further. In such a case, operate the respective HOME switch on the operator's console to set it back to the HOME position.

2） Snag Protecting Function

Snag is a condition that typically occurs when a lifting spreader hooks or hits an obstruction such as the cell guides or hatch combings in the ship's cargo hold.

As snag occurs, the ropes' tension will increase rapidly and the hydraulic pressure of the snag cylinders increases correspondingly. When the pressure of any one of the four cylinders rises to the preset value of its pressure switch, a hardwired (fast) signal is sent directly to the hoist drives that initiates an emergency stop to the hoist motors. At the same time, the relief valve will open to blow the fluid and the cylinders will retract.

After the snag occurs, the snag indicating light will be on in the operator's cab, an audible warning will be sounded and the control power will be shut off. Meanwhile, only lowering the hoist is allowed at the reduced speed to release the strain on the ropes and no hoist raise motion is permitted.

In order to exit the snag condition, first the operator must reset control power and lower the hoist to remove the load (hoisting is not permitted). The crane is then to be inspected for damage before the SNAG RESET key switch, located in the E Room is reset. Once reset full crane functionality is permitted.

1.3.2 Boom hoist system

1.3.2.1 Boom hoist drive

The boom hoist drive is located in the machinery house. It is powered by a variable frequency AC motor connected to the reducer input shafts through a flexible elastomeric type coupling (high speed). A rope drum is connected to the output shaft of the reducer through the flexible gear coupling (lower speed). The driving force of this equipment is transmitted from the motor to the drum by the high speed coupling and drum coupling and amplified through the reducer driving the drum and rope to raise or lower the boom. The data of main components as following:

Table 1-3: Main Data of Major Components for Boom Hoist Drive

Motor	Type	BDK-1KMU(YASKAWA)	Drum Brake	Type	SB315
	Power	375kw		Brake Torque	2X254.5kNm
	Rotating speed	1750rpm	Motor Coupling	Type	AFN SB100-560x30
Reducer	Type	ML4PSF120X(SEW)		Rated Torque	4950Nm
	Ratio	100.56		Brake Torque	9900Nm
	Rated Output torque	329KNm	Drum Coupling	Type	DC10B
Motor Brake	Type	YP31-ED2000-560x30		Rated Torque	560kNm
	Braking Torque	5750Nm	Wire Rope	Diameter	Φ35.5mm

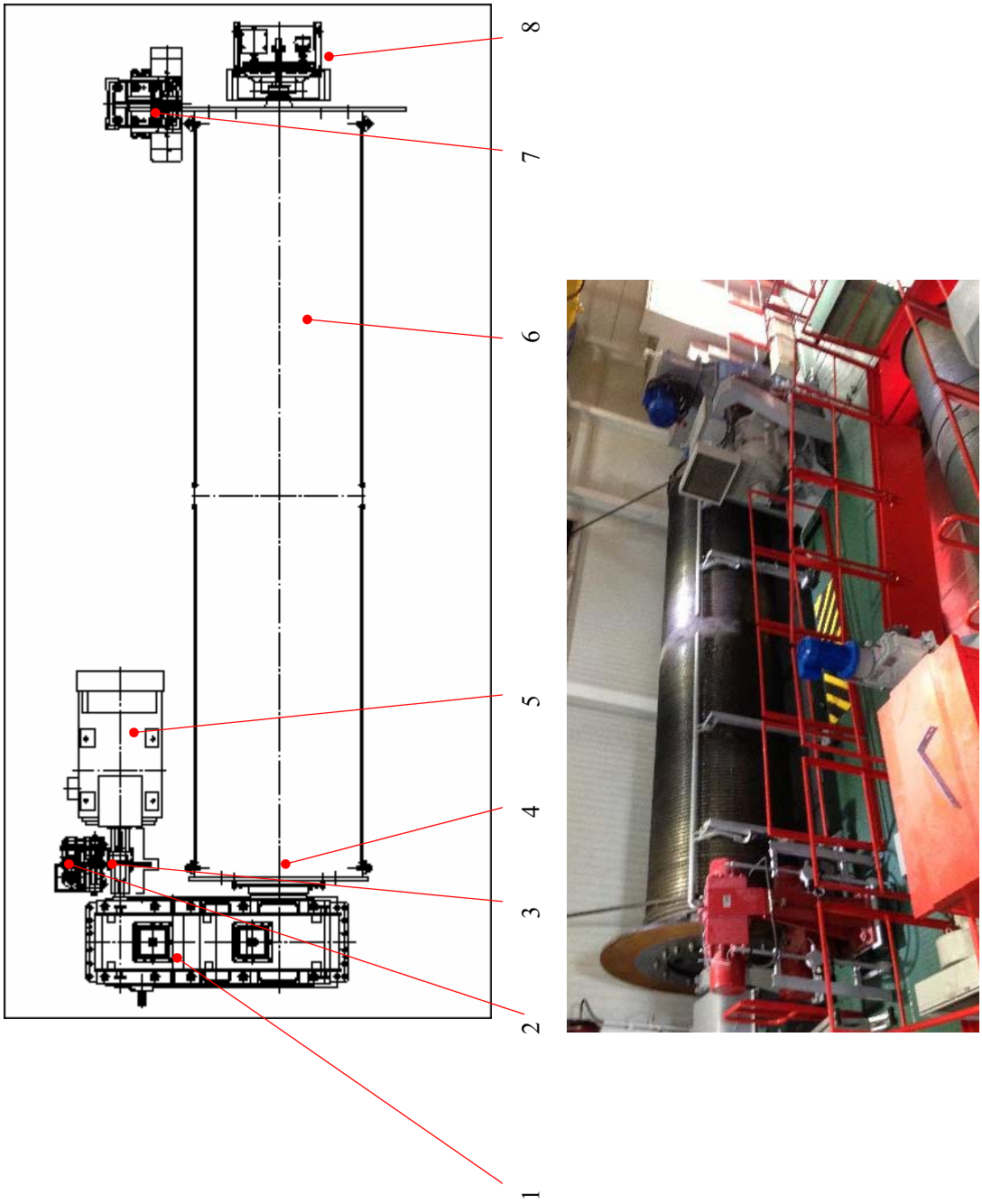


Figure 1-7 Scheme and Photograph for Boom Hoist Drive

1 Reducer
2 Motor Brake
3 Motor Coupling
4 Drum coupling
5 Motor
6 Drum
7 Drum Brake
8 Bearing Seat and Limit Switch

An emergency drive is located on the top of main reducer, its motor flanged onto the cycloidal planetary pin wheel speed reducer. Upon failure of the boom hoist main motor or drive/control system and can not be fixed immediately, this device can be used to raise or lower the boom at low speed. In case the crane loss of high voltage power, it is also capable of slowly raising or lowering the boom by plugged on shore power.

1.3.2.2 Boom hoist rope reeving system

The wire rope pays out from the drum, then runs through the following route: Boom hoist drum → Sheave block on top of A-frame → Inner single sheave on the boom → Sheave block on top of A-frame → Outer sheave blocks (4) on boom, rope runs 8 times to and fro between sheave block on top of sheave and outer sheave block on boom for either side reeving as shown in Figure 1-15.

The hoisting or lowering of the boom can be altered by changing the rotating direction of the motor. The boom hoist drive lifts the boom to the 45º stowed position or to the 80º maintenance position through the reeving rope. The boom hoist reeving includes two independent and equalized sets of falls. Either of them can catch the boom in the event of failure of one set. All sheaves contained with bearings and through covers are designed as a unit when removed or installed.

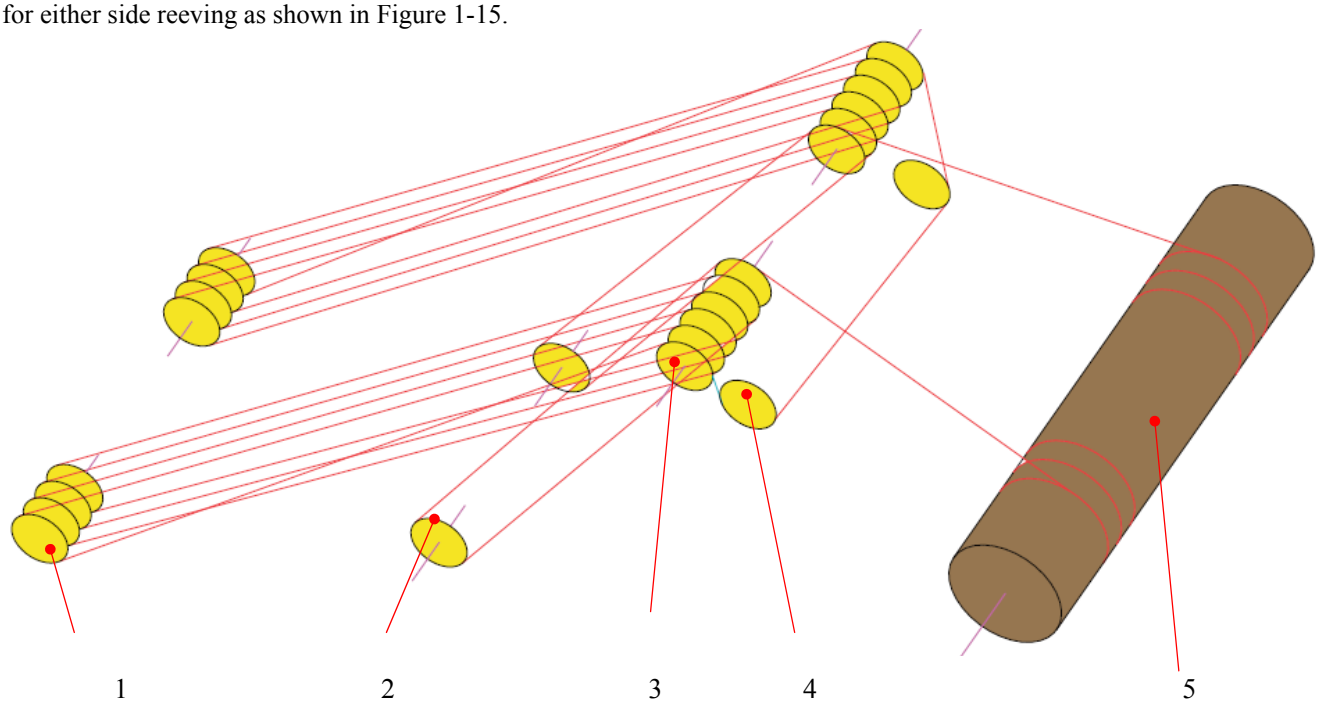


Figure 1-85 Scheme of Boom Hoist Rope Reeving System

1 Outer Sheave Blocks On Boom 2 Inner Single Sheave on Boom 3 Sheave Blocks on Top of the A-Frame
4 Equalizer Sheave on Top of the A-Frame 5 Boom Hoist Drum

1.3.2.3 Boom hoist interlocks

1) Over-speed switch
As shown in Figure 1-16, the external over-speed protection for the boom is provided at the end of the drum. Actuation of switch initiates an emergency stop. The theoretical trip speed setting for the switch is 110% of the motor maximum speed. In order to resume the crane operation, the switch must be manually reset from its back once it is tripped. (Refer to Figure 1-16) As an optional device, which is connected with the motor shaft end, the incremental encoder can transmit the real-time speed signal then PLC will evaluate over-speed or not. (Refer to Figure 1-17)

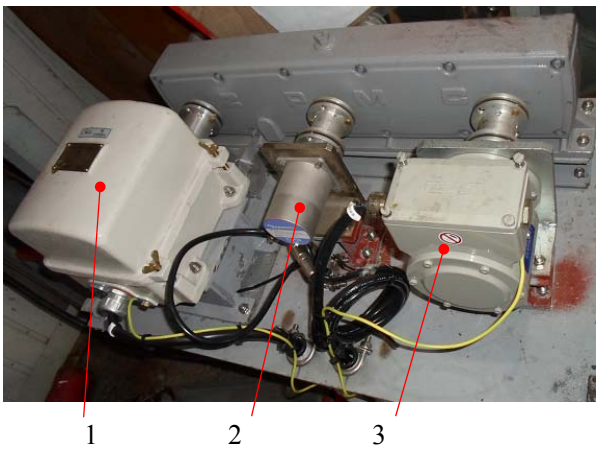


Figure 1-16 Limit Switches at Drum Shaft end

1 Cam Switch 2 Absolute Encoder 3 Over-speed Switch



Figure 1-17 Incremental Encoder

2) Motor brake limit switch
The motor brake is monitored by a brake released limit switch (one normally open contact) and a brake manually released limit switch (one normally closed contact). Under no command of boom up or down, the boom motion is inhibited from operating as long as either switch is activated.

3） Drum brake limit switches
Two boom drum brakes are monitored by a brake released limit switch (one normally open contact). Under no command of boom up or down, the boom motion is inhibited from operating as long as either switch is activated.

4） Motor thermal switch
The motor features an embedded over-temperature switch (one normally closed contact). Actuation of the switch indicates the motor temperature is at fault level and the boom motion is inhibited.

5） Trolley park position check
The trolley park position is monitored by a magnet-operated proximity switch. The boom motion is inhibited from operating as long as the trolley is not indicated to be in the park position.

6） Boom stowage pin stored in storage socket check
The boom stow pins are monitored by proximity switches. Actuation of the switch indicates that the stow pin is rested in the storage socket and boom motion is permissive. Boom motion is inhibited from operation as long as either of switches is deactivated.



Figure 1-18 Limit switch for checking stow pin in latch

7） Rope slack protection
The boom ropes are monitored by two lever operated limit switches, one for each of the rope falls. The boom motion is inhibited from operation (only lowering direction) as long as either of the switches is actuated.



Figure 1-19 Rope slack switch

8） Rope equalizing
The midpoint of the rope between the two equalizing sheaves is secured by a clamp. As either set of rope fall broken, the clamp will turn into the new dead end. And it will be caught by the restricting structure on the top of apex so as to prevent the accident of boom falling. A proximate limit switch will be settled on each end of the clamp to check the rope equalizing between the two sets of falls.

1.3.2.4 Boom traveling limits

Normal boom hoist limits are controlled by the absolute encoder mounted on the drum shaft end. The home position of the boom hoist is set by lowering the boom to down stop position.

In addition to the absolute encoder limits, there is a lever operated limit switch mounted on the top of A-frame (As shown in Figure 1-20) and a multi-cam rotary limit switch mounted on the end of boom drum (As shown in Figure 1-16). This redundant design provides the over travel protection in a better way.



Figure 1-20 Mechanical Limit switches for Boom stowage

1） Raise over travel stop
As mentioned above, it is under the control of the lever operated limit switch mounted on the top of the A-frame together with the multi-cam rotary limit switch and an absolute encoder. Actuation any of these limits initiates an emergency stop to the boom hoist motor. The “limit back-out bypass” selector switch in the electrical house permits boom lowering down.

2） Raise normal end stop (maintenance position at 80°)
Under control of the lever operated limit switch mounted on the apex top together with the multi-cam rotary limit switch and an absolute encoder, actuation any of these limit switches initiates a ramp stop and functions as a synchronization point for the boom positioning system.

3） Raise slowdown check (maintenance position at 80°)
Cam limit switch actuates at the check point. Wired as digital input of PLC to the overriding controller, the check point is set at ±5% of slowdown starting speed.

4） Raise slowdown (maintenance position at 80°)
Controlled by the absolute encoder together with the multi-cam rotary limit switch, the speed check point of slowdown is the start of the normal slowdown.

5） Lower slowdown
This is Similar to the raise normal slowdown.

6） Lower slowdown check
This is Similar to the raise slowdown check.

7） Lower normal end stop/boom level
Cam limit switch actuates above the synchronization point. As the digital input from PLC to the overriding controller, the switch functions as the homing point (zero) of the absolute encoder.

8） Lower over travel verification
Cam limit switch provides mechanical backup for over lowering.

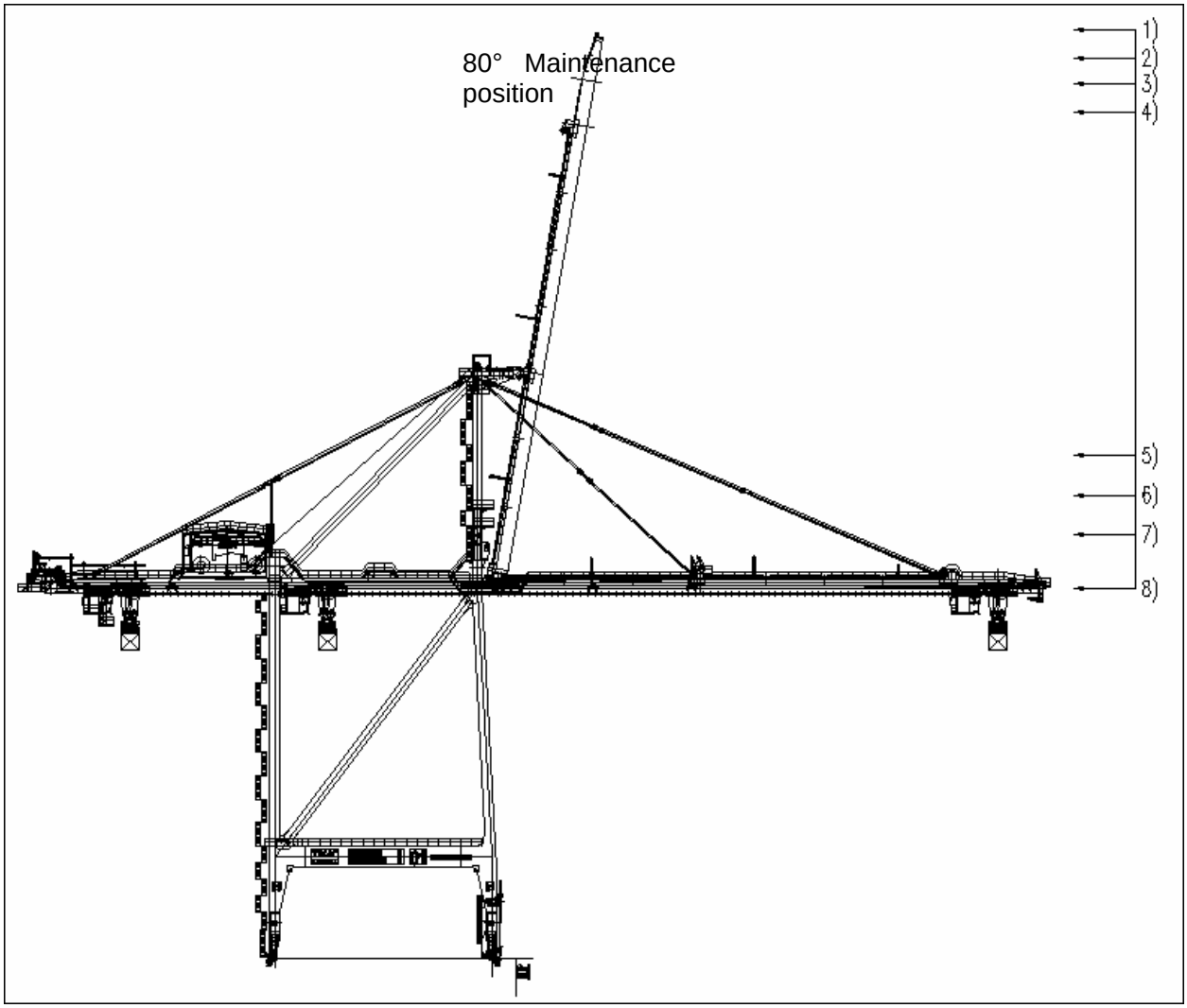


Figure 1-21 Scheme for Boom Traveling Limits

1.3.3 Trolley travel system

1.3.3.1 Trolley travel drive

The trolley travel drive is located in the machinery house. It is powered by a variable frequency AC motor connected to the reducer input shaft through a flexible elastomeric type coupling (high speed). A rope drum is connected to the output shaft of the reducer through the flexible gear coupling (lower speed). The driving force of this equipment is transmitted from the motor to the drum by the high speed coupling and drum coupling amplified through the reducer driving the drum and rope to propel the trolley forward (seaside) or backward (landside). The data of main components refers to Table 1-4.

Table 1-4: Main Data of Major Components for Trolley Travel Drive

Motor	Type	BDK-1KMU	Drum Brake	Type	
	Power	275kw		Brake Torque	
	Rotating speed	1750rpm	Motor Coupling	Type	MLPK10-560x30
Reducer	Type	FH955.21.A6A-00		Rated Torque	4.5Nm
	Ratio	20.702		Brake Torque	11.94 kNm
	Rated Output torque	75kNm	Drum Coupling	Type	DC05B
Motor Brake	Type	YP31-ED1250-560x30		Rated Torque	125kNm
	Braking Torque	3.55kNm	Wire Rope	Diameter	Φ25mm

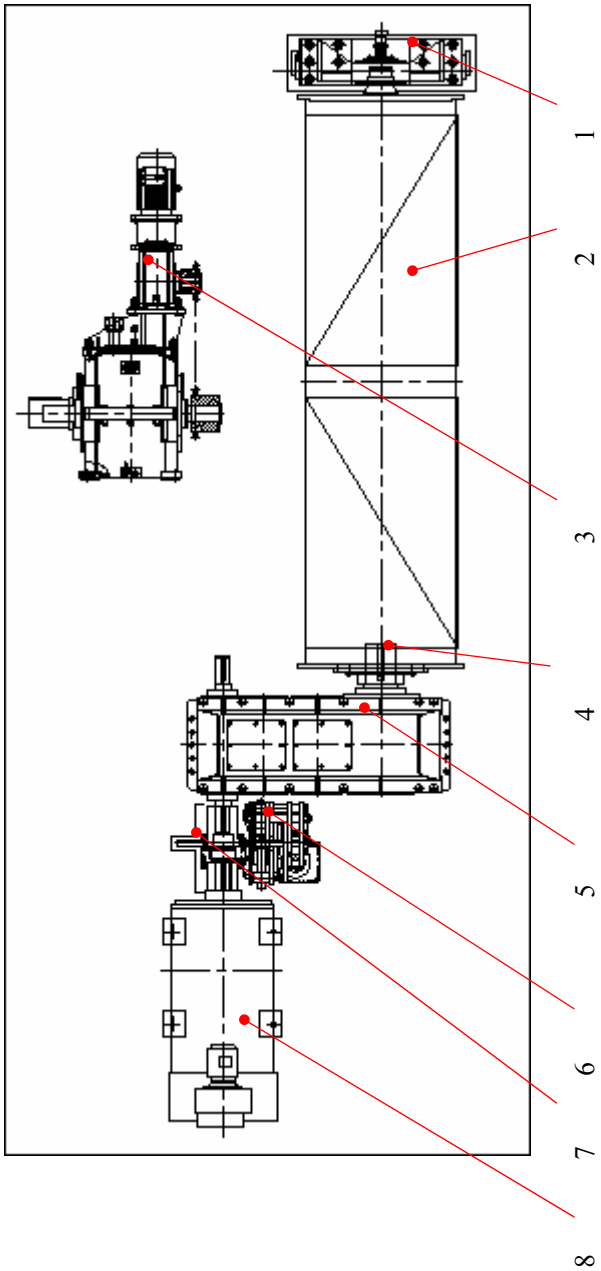




Figure 1-22Scheme and Photograph for Trolley Travel Drive

1 Bearing Seat
2 Drum
3 Emergency Device
4 Drum Coupling
5 Reducer
6 Motor Brake
7 Motor Coupling
8 Motor

The trolley travel drive is equipped with an emergency drive. The device is capable of moving the trolley slowly forward or backward powered by the shore power in case the trolley motor or drive control system failed and can not be fixed immediately.

1.3.3.2 Trolley rope reeving system

The landside wire rope starts from the drum then run through the following route: Drum → Sheave of rope tensioning device at girder rear section → Trolley equalized sheave→ Rope clamp → Trolley equalized sheave →Sheave of rope tensioning→Drum. The seaside wire rope starts from the drum then run through the following route: Drum → Boom tip trolley sheave → Trolley equalized sheave→ Rope clamp → Trolley equalized sheave → Boom tip trolley sheave→Drum.

Wrapping around the trolley drum, the four ends of the two wire ropes are clamped on the drums. Both middle points of the trolley ropes are secured on the trolley,which creates two ropes at the waterside of the trolley and two ropes at the landside of the trolley. They tow the trolley backward and forward.

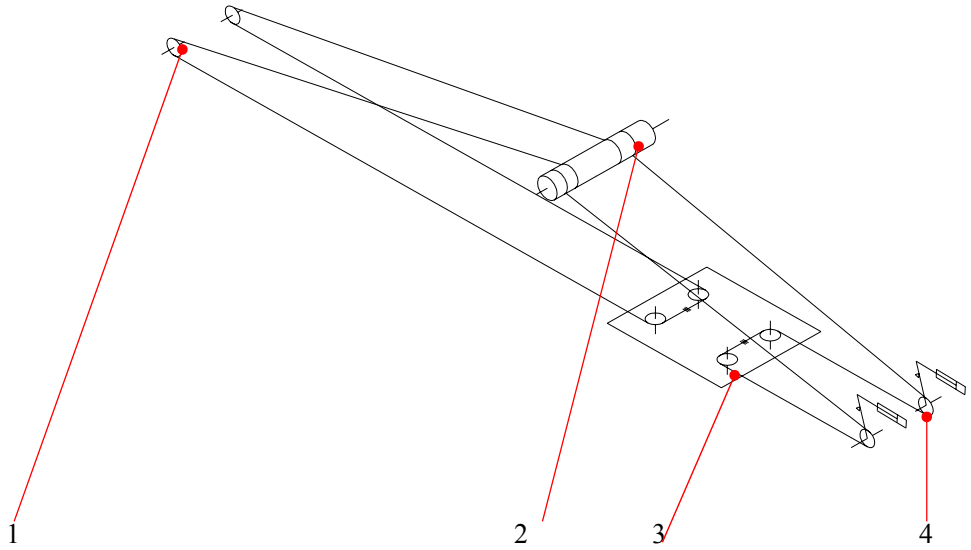


Figure 1-23 Trolley Rope Reeving System

1 Boom tip sheave assembly 2 Trolley drum 3 sheave assembly on the trolley
4 Sheave of rope tensioning device at girder rear section

1.3.3.3 Trolley interlocks limits

1） Motor brake limit switch
The brake is monitored by a brake released limit switch (one normally open contact) and a brake manually released limit switch (one normally closed contact). Under no trolley command from joystick, the trolley motion is inhibited from operating as long as the switch is activated.

2） Motor thermal switch
The trolley motor features an embedded over-temperature switch (one normally closed contact). Actuation of the switch indicates the motor temperature is at fault level and initiates a ramp stop. The trolley motion is inhibited from operating as long as the switch is opened.

3） Boom hoist interlocks
If the boom is not at right position (not fully lowered to level or fully up to stop position), the trolley motion will be disabled.

4） Cab access gate
Each of the two gates to the trolley, one on the portal frame and the other located on the trolley, is monitored by a gate closed proximity switch. Actuation of either of the switches indicates the respective gate is not closed well the trolley motion is to be inhibited.

5） Trolley rope tensioning limit switches
This protection consists of four proximity switches, two for each utmost stroke of the rope tensioning cylinders. The trolley motion is inhibited from operation as long as either of them is activated.

1.3.3.4 Trolley traveling limits

1） Waterside over travel stop
Controlled by Lever operated limit switch together with the absolute encoder mounted on trolley drum shaft end, actuation any of these limit initiates an emergency stop to boom hoist motor. The “limit back-out bypass” selector switch in the electrical house permits trolley traveling towards landside.

2） Waterside normal end stop
Controlled by proximate limit switch together with the absolute encoder, actuation of this normal working end stop initiates a ramp stop.

3） Waterside slowdown check

Controlled by proximate limit switch together with the incremental encoder attached on the motor end shaft, actuation of the proximate switch and trolley speed verified not slowdown to set range by the encoder will initiate an emergency stop.

4） Waterside slowdown check
Controlled by proximate limit switch together with the incremental encoder attached on the motor end shaft, actuation of the proximate switch and trolley speed verified not slowdown to set range by the encoder will initiate an emergency stop.

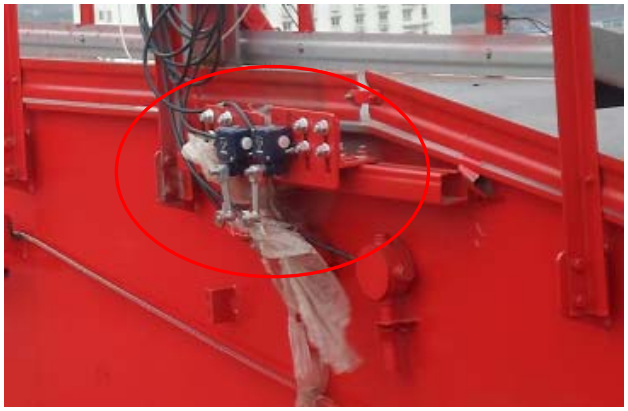


Figure 1-24: Roller Lever Operated and Proximate Limit Switches around Boom End.

5） Trolley Encoder Reset
Actuation of the proximate switch initiates resetting a new position value for the absolute encoder.

6） Waterside normal end stop (boom stow position)
Actuation anyone of the proximate switch together with absolute encoder indicates a ramp stop for the trolley.

7） Park position
Controlled by magnet operated proximity switch together with the absolute encoder, actuation of this limit switches indicates trolley at park position and permits boom motions.



Figure 1-26: Trolley Traveling Position Switch

8） Sill beam protection
Under the control of the absolute encoder, actuation of any sill beam stop initiates a ramp stop in order that the spreader safely crosses over the sill beam. When hoist lowering within zone H2-H3 as shown in Figure 1-27, trolley position will be checked whether it is at the preset area or not, and trolley will be stop if it is .

9） Landside smart slowdown
This is similar to the waterside smart slowdown.

10） Landside slowdown check
This is similar to the waterside slowdown check.

11） Landside normal end stop
This is similar to the waterside normal end stop.

12） Landside over travel stop
This is similar to the waterside over travel stop.

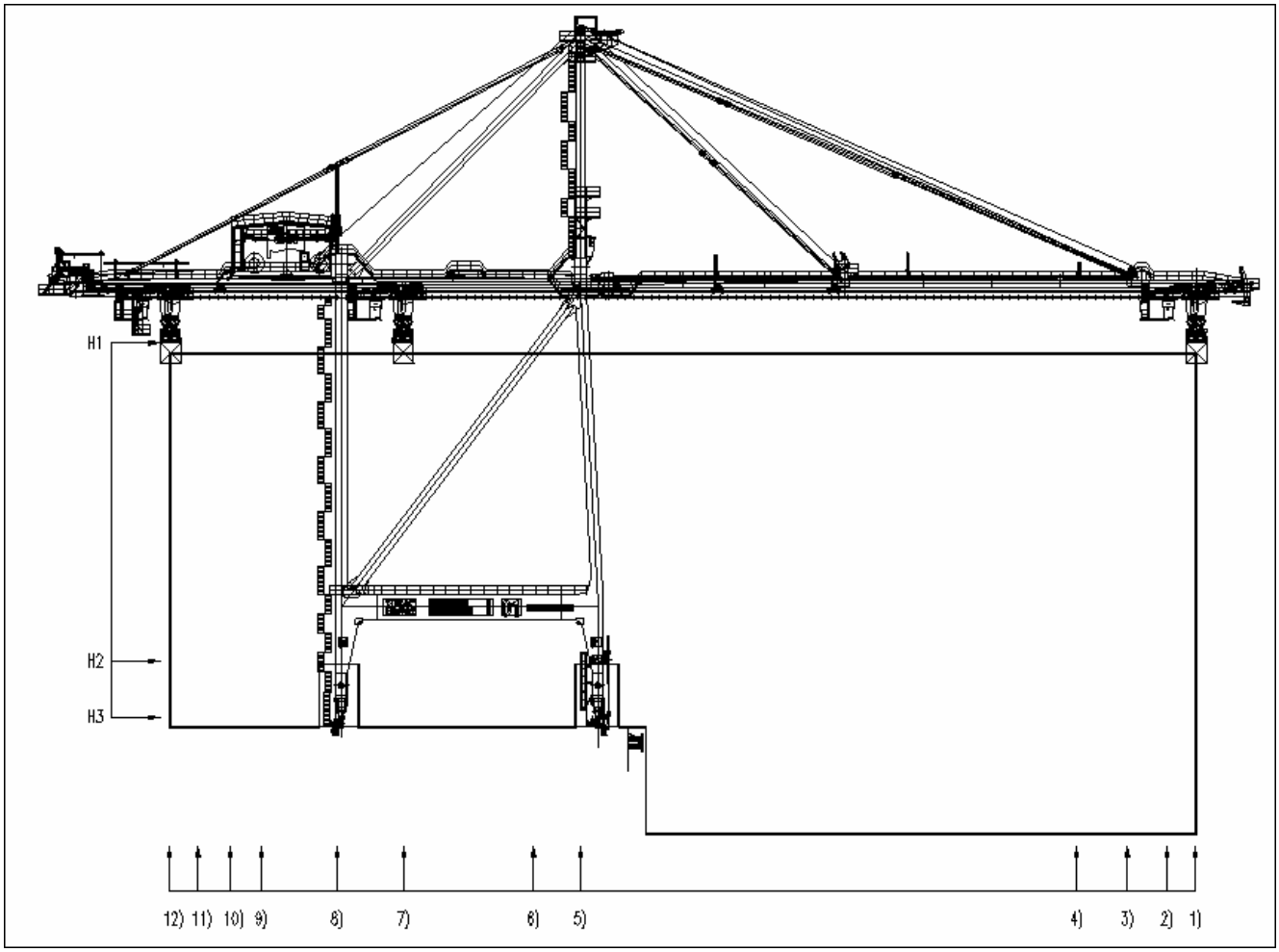


Figure 1-27 Trolley Traveling Limit Switches Arrangement

1.3.3.5 Trolley travel rope tensioner

Trolley rope tensioners are located at the rear of the girder and apply appropriate tension to the trolley travel ropes by actuating two hydraulic cylinders attached to trolley sheave through the sheave arm. With this function, surging of the trolley due to the loosening of the wire rope during loading or unloading of the container will be minimized.

	In many cases, the wire rope length needs adjustment due to the stretching and slackening of the wire rope, and this variation is difficult to notice especially in continuous catenary roller support system.



Figure 1-28 Trolley Travel Rope Tensioner

The cylinders are powered by the multi-function power unit with pre-set pressures to the hydraulic cylinders. The rope tensioning cylinders will retract to prevent the ropes from over-tensioning due to the changes of the waterside rope length as the boom rises. The cylinders will extend to keep the ropes under proper tensioning due to new ropes stretch after a period of operation. For more information on the hydraulic system, please refer to Chapter 5 in this manual and Hydaulic Maintenance

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The position of the cylinder is monitored by two proximity switches and actual position is decided by the real presetting pressure / tension force. The desired position of the cylinder is that the sheave arm is almost vertical under the pressure of hydraulic system. The arm may offset from its original position and trolley motion is allowed until either of limit switches is activated. If limit switch tripped and cylinders in normal pressure wire rope length needs to be adjusted.

1.3.3.6 Trolley

The seaside wire rope starts from the drum then run through deflecting sheave on boom tip back to trolley with its midpoint secured on a screw rod block used to adjust wire rope length conveniently in some range.

Four traveling wheels are on the trolley frame corners. The wheels have anti-friction bearings with split pillow blocks for easy removal. There are altogether four trolley trucks mounting on the trolley structure. Each corner has one wheel Each wheel has one spherical bearing for aligning the wheel.

Eight main hoist rope sheaves are mounted on the trolley frame. They accomplish the rope pickup and diverting.

The trolley frame has four pneumatic-hydraulic buffers at the seaside and the landside to minimize the impact on the end stop in case of an electrical control failure.

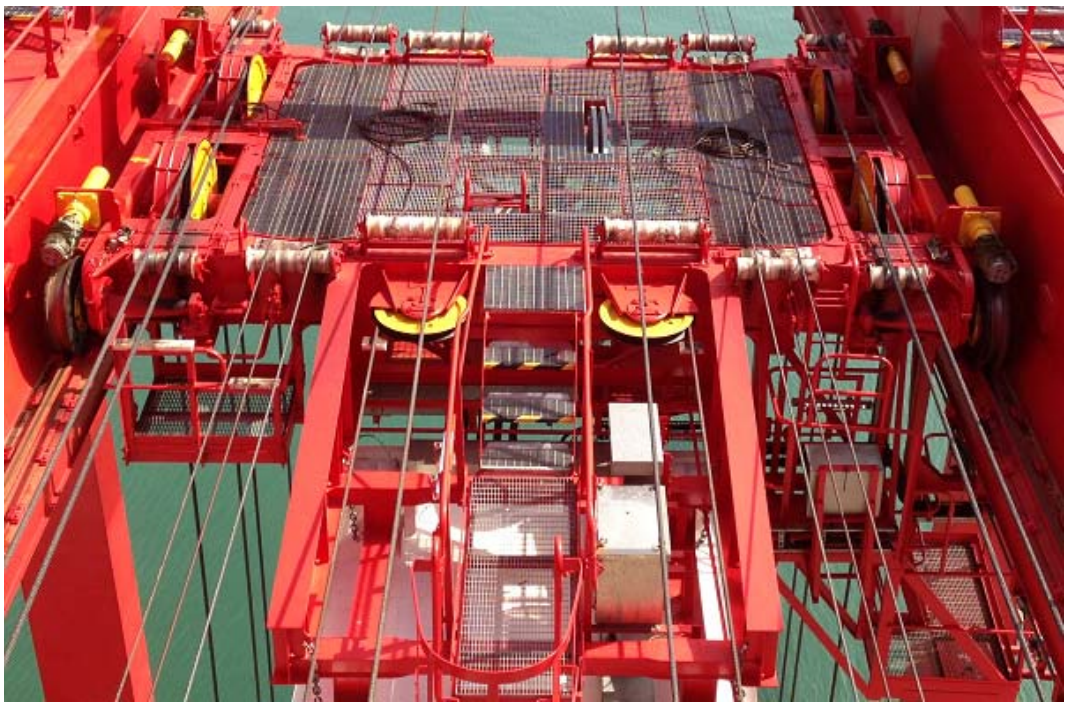


Figure 1-29 Trolley Assembly Photograph



Figure 1-30 Rope clamp on the trolley frame

1.3.3.7 Operator's cab

The cab is purposefully and ergonomically designed to serve the crane operator. Attention was focused on providing comfort and convenience so that the operators can concentrate on the serious task of handling containers, and not be distracted or encumbered by uncomfortable surroundings.



Figure 1-31 Operator's Cab Layout

All primary crane functions can be achieved in the cab. The cab is mounted to the trolley framework on its landside to provide the best possible view during loading or unloading operation.



Figure 1-32 Operator's Cab Appearance



Figure 1-33 In View from Floor Window in Operator's Cab

A seat with safety belts and shoulder harness, adjustable to accommodate a person's weight, height and distance from the front windows, is positioned so that the operator can reach all control switches comfortably.

Soft fluorescent lighting can brighten the interior of the cab for reading, or the operator can use the dimmer mounted alongside the left console to darken the cab or reduce lighting to your desired intensity.

All master switches, indicating lights, selector switches, joy sticks, and other control devices have been conveniently arranged on the left and right consoles, the auxiliary panel, twist lock indicating panel along the foot rest, and on the three computer and TV monitors, so the operator can access any of them without leaving your chair.

1.3.4 Gantry

1.3.4.1 Gantry travel drive

Gantry traveling devices drive the crane travel along the runway. Each corner consists of 8 wheels which half are driving wheels. Sixteen AC variable frequency motors power the gantry travel. Two AC drives are used, one for the waterside motors and the other for the landside motors. Speed coordination between the two motions is controlled through a pulse tachometer mounted on each far left motor. Since motors are numbered 1 through 16 starting at waterside right and progressing counterclockwise motor number 16 is at landside right. To prevent the crane from shifting in storm wind (stowed wind), two sets of stowage pins are provided under the seaside and landside sill beams. In addition to motor mounted brakes which brake the driving wheels, hydraulic rail brakes are mounted at each upper equalizer beam to prevent the crane from moving in wind gusts during the operation.

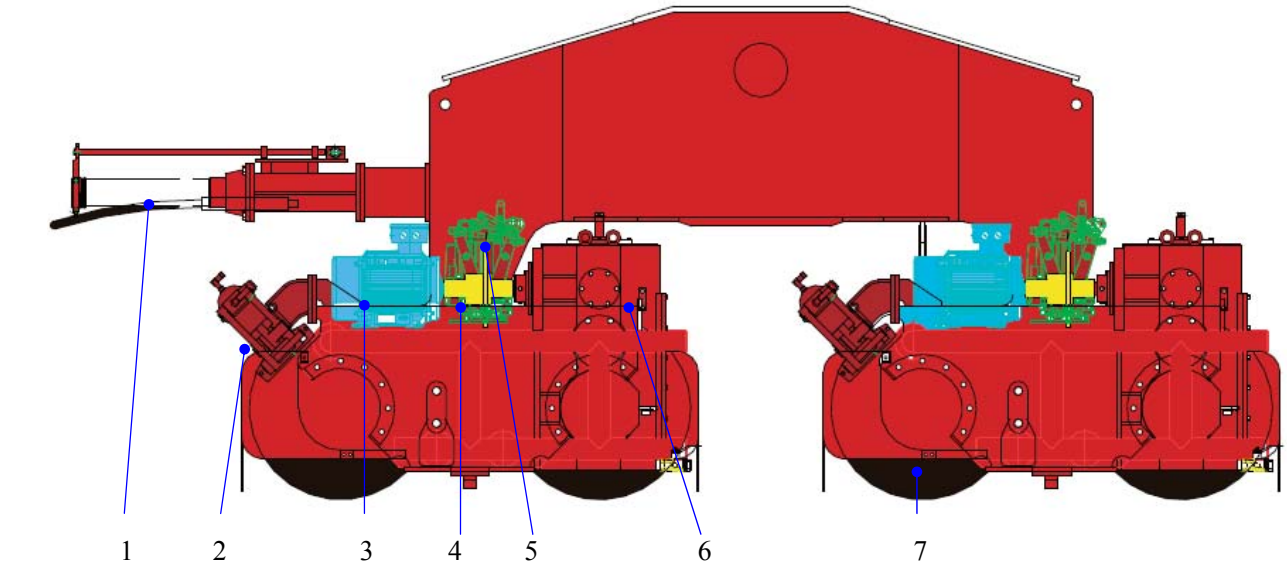
The driving force from the motor is transmitted through the gear coupling and gear reducer to the drive wheel. Travel direction of the gantry can be altered by changing the rotating direction of the motor. A electrically released brake for each drive unit is provided inside the motor. The total 16 motor brakes are sufficient to stop the gantry with wind speed of 25m/sec. The motor brakes combined with the rail brakes are sufficient to hold the crane at the wind speed 35m/sec. But if the forecasted wind speed is over 35m/sec, the stowage pin is required to be engaged in proper way. The data of main components refers to table 1-5.

Table 1-5: Main Data of Major Components for Gantry Travel Drive

Motor	Type	FEK C21 OU	Rail Brake	Type	JLQ115.00-KH
	Power	20kw		Contact pressure	115KN
	Rotating speed	1750rpm	Motor Coupling	Type	MLPK06-315x20
				Rated Torque	630Nm
Reducer	Type	FKR460.96.D1B-00	Buffer	Type	7/500BQD140-517
	Ratio	96.734		Max Permissible Impact force	700KN
Motor Brake	Type	YP11-315x20-ED220/50		Stroke	500mm
	Braking Torque	260Nm	Wheel	Diameter	Φ800mm



Figure 1-34: Seaside Gantry full view





1 Buffer
2 Rail Brake
3 Motor
4 Motor Coupling
5 Motor Brake
6 Gear Reducer
7 Gantry Wheel

Figure 1-35: Scheme & Photography of Gantry Drive Unit

1.3.4.2 Interlocks for Gantry Drive

1） Motor brake limit switches
Each one of the 16 motor brakes is monitored by a proximate limit switch.

2） Rail brake limit switches
Each one of the rail brakes is monitored by a proximate limit switch.

3） Gantry stow pins
Each of the two stow pins on gantry, in both landside and waterside, is monitored and checked whether in released position by a proximity switch. As long as either pin is not in the retracted position, the gantry motion is inhibited from operating.

4） Gantry buffer limit switch
Each of the two gantry buffers located at landside or waterside is monitored by a proximity switch. The gantry motion is inhibited from operating and a warning sounded as long as either side buffer is compressed to setting position.



Figure 1-36: Gantry Buffer Limit Switches

5） Gantry Ultrasonic Sensor
The scanner device mounted on the small equalizer beam end consists of an Ultrasonic Sensor and a reflector. Actuation of this scanner initiates an emergency stop for gantry if obstacles are detected along the gantry traveling direction.

6） Anemometer (wind speed) system
Wind speed and direction monitoring for the gantry drive is provided by anemometer system. The anemometer transducer located on top of the A-frame, for measuring the wind speed and direction, a control module located in the electrical room, and an digital meter located in the operator's cab for displaying wind speed.

The anemometer has two set points. If greater than the warning level of 25m/sec for a period greater than 10 seconds, an amber light will illuminate in the operator's cab and a warning signal will sound through the PA system audible in the operator's cab, machinery house and ground level. The crane operator should prepare to gantry and stow the crane.

A second alarm on the anemometer is set to sound at 28m/sec. This alarm has a different tone and the power for gantry traveling. The power will be disconnected and the gantry operation is inhibited.

1.3.4.3 Gantry Maintenance Cab

There is a gantry maintenance cab under the landside sill beam. The cab is provided with electric lock. Only those who have the mechanical key or the electronic key can access this cab. A ground control station and CMS terminal are mounted in this cab. This control station will help to maintain the spreader and stow the crane.

All functions of the spreader can be achieved on this station. The main hoist can be operated with empty spreader only.



Figure 1-38: Gantry Maintenance Cab

1.4 Crane safety facilities

1.4.1 Crane Stowage Device and Tie-Down Device

In this project, each stowage plate with a hole can fasten the crane on the dock.



Figure 1-39: Gantry Stowage Device

There are totally two crane stowage devices (plates) that are mounted near the crane centerline under both the waterside and landside sill beams. The stowage pins can be raised or lowered. When raised, the stowage device is held by a catch.

It is important and necessary to properly stow the crane safely on the designated site of dock (stowage pit). If not stowed, there will be a danger in the event of a sudden high wind condition. The motor brake and wheel brake can only guarantee locking the wheel with no rotation but can not prevent the crane from sliding along the rail. Once in motion, a crane can develop momentum and strike another crane or other obstacle and cause serious damage.

	**No operation is allowed when wind exceeds 20m/sec! Crane must be stowed at its stowage position. Crane must be secured in Stowage conditon when wind exceeds 30m/sec.**



Figure 1-40: Gantry Stowage Device Located under sill beam

1.4.2 Boom stowage device

This device is for securing the boom when the boom rises to the maintenance position (80°). It consists of two sets of pins and sockets, as shown in Figure 1-41. The pins are stored in the storage sockets under normal operation. The boom motion is inhibited from operation as long as either of the pins is not inside the storage sockets for safety.

The boom motion will ramp stop when the normal stop limit switch is activated. At that time, the latch hole on the boom is aligned with that on the stowage device. Pull pins out of the storage sockets and insert them into the latch hole. Insertion of an anchor pins must be done by manpower.

When trying to lower the boom, the pins must be inserted back into the storage sockets. Otherwise there is no motion of the boom. There are limit switches to sense the pins.



Figure 1-41: Boom Stowage device

1.4.3 Ship Collision Protection

A stainless steel trip wire runs down each side of the boom to protect the lowest surface of the boom from contacting the ship's structure. A proximity limit switch is located at the end of the steel wire on both sides of the boom. If either limit switch is activated, further travel toward ship will be inhibited and a warning signal will be sounded through the public address (PA) system. But the crane can gantry away from the ship and the limit switch is de-activated. A bypass switch in the cab allows the gantry to travel in the detected direction. In this bypass mode, the warning light and the horn will activate.



Figure 1-42: Ship Anti-collision Device

1.4.4 Obstruction lights

Each location of boom tip, trolley girder rear end and top of A-frame is equipped with an obstruction light to flash at night for warning the flying objects avoiding.



Figure 1-43: Obstruction Light

1.5 Auxiliary equipments on crane

1.5.1 Catenary trolley System

The purpose of the catenary trolley system is to support the main hoist and trolley towing ropes and minimize the rope sag and bouncing during container handling operation.



Figure 1-44:Catenary trolley system



Figure 1-45: Sheaves on Boom Tip

1.5.2 Power cable reel

A cable reel device rides on the frame link beam to delivery the high voltage power for the crane. As crane traveling, the cable is wound or paid out by the cable reel, which is driven by the gearbox output torque transmitted from the motor through a coupling. A guide roller device above the ground helps to lay the cable in a pit or pick up from it. On the opposite side of gearbox, a high voltage slip rings connected with the output shaft feeds H.V power to the crane.



Figure 1-46: Power Cable Reel System



Figure 1-47: Inside View of Slip Ring Cabinet

1.5.3 Slip contact system

Slip contact System is the power cable and control cable carrying device traveling with the trolley.



Figure 1-48: Slip contact system

1.6 Maintenance equipment

The maintenance equipments provided on the crane includes service crane inside machinery house, rope re-reeving device, service platform and tool kit, etc,.

1.6.1 Machinery house service crane

A service crane with 12 tons rated capability is provided in the machinery house. Three motions of hoisting, transverse crossing and longitudinal traveling can cover the whole house. The service crane spans the machinery house and is capable of centering over, lifting and traveling with any component inside the machinery house as well as carrying components between the machinery house and wharf level. In order to position parts accurately, it is furnished with two stage hoisting speeds, one for fast and one for slow.



Figure 1-49: Service Crane in Machinery House

1.6.2 Service crane outdoors

Two type of service cranes are provided outdoors on this crane, sample crane and over-head service crane. A sample crane located on boom tip and one on the top of A-frame serve for replacing sheaves. On the girder end, an over-head service crane is supplied to take care of the cylinders, sheaves on rear tie beam as well the sheaves and wheels on trolley. Some handling tools such as plain trolley and chain blocks need to be put on temporarily while moving components during maintenance.



Figure 1-51: Sample Crane on Boom tip



Figure 1-52: Overhead Service Crane on Girder End

1.6.3 Wire Rope Re-reeving Device

A re-reeving device is mounted over the main hoist in the machinery house. It consists of two cable drums including two motorized devices equipped with brakes. During re-reeving procedure, the device winds the old rope on its take-up reel tightly and the new wire rope is on the feeder reel. For detailed rope re-reeving procedure, please refer to Chapter 4 of this manual.



Figure 1-53: Re-Reeving Device Mounted Over the main hoist Drive

1.6.4 Tool Kit and Workbench

Totally two sets of tool kits and one workbench are provided on the crane and stored in the machinery house for maintenance personnel



Figure 1-54: Tool Kits in the Machinery house



Figure 1-55: Workbench in the Machinery house

1.7 Walkways, access and elevator

1.7.1 Walkways and Platform

Each crane access to every part of the crane needing routine inspection or maintenance is provided with stairways, walkways, and ladders. The main access stairway is located on the landside left leg. Rest platforms are provided at various elevations along the stairways which can be used to access the portal tie beam, operator's cab parking position, or machinery house elevation.

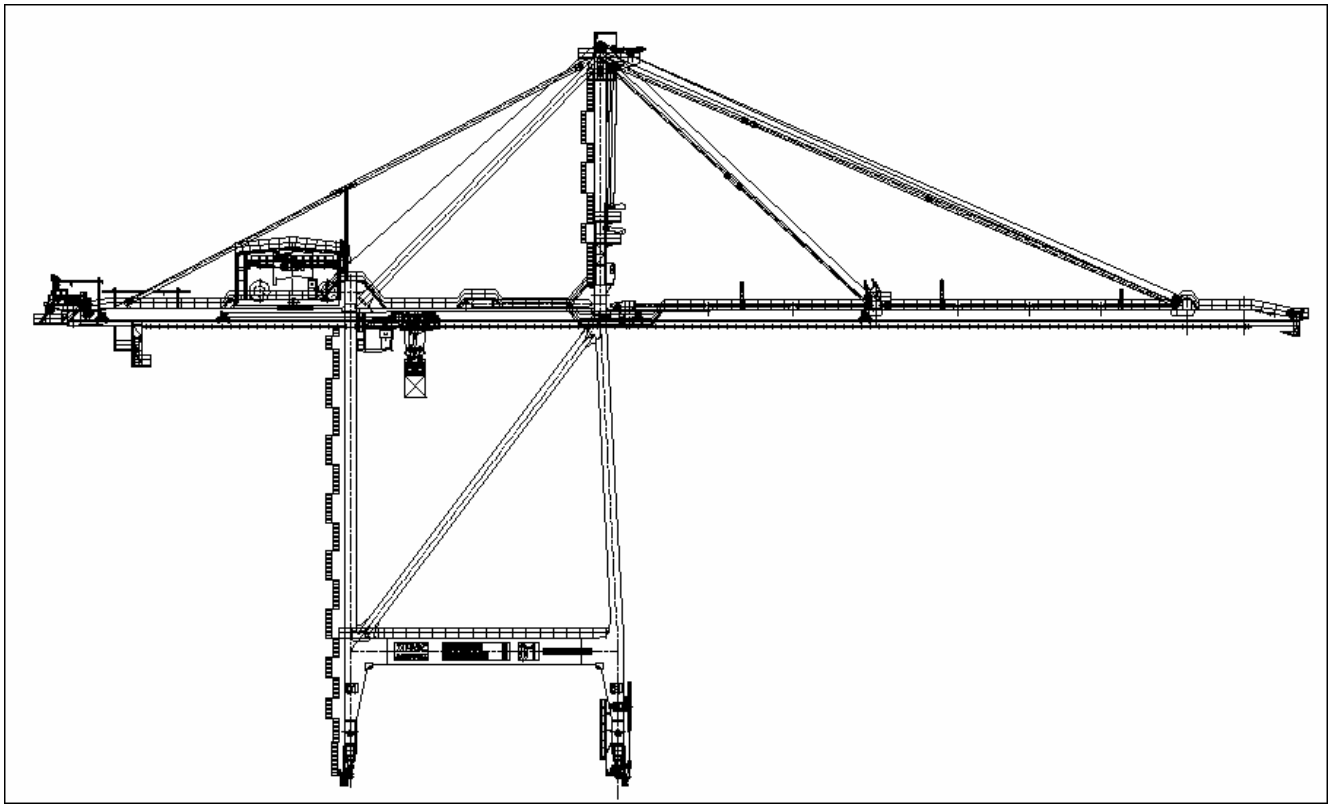


Figure 1-56: Side View of Access General Arrangement

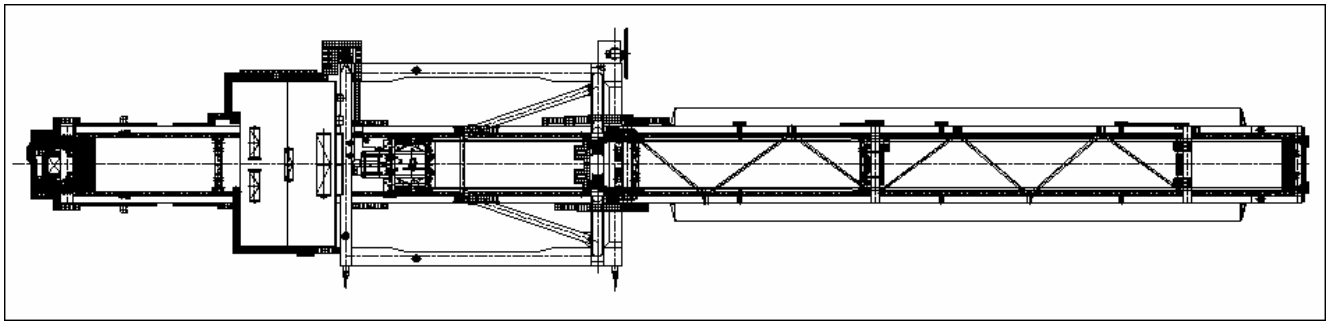


Figure 1-57: Top View of Access General Arrangement



Figure 1-58: Maintenance Platform for Boom Fore-stays

1.7.2 Elevator

On the same leg which the main access located at (crane stairway access), a man-carried elevator is installed for up and down. The main data of the elevator refers to table below. For detailed maintenance instruction, please refer to an independent manual.

Table 1-6: Main Data for Elevator

Rated Capacity	400kg/4 人	Care Size (inside)	1000x1000x2150	
Driven Type	Rack & Pinion	Car Floor Area	1 square meter	
Voltage	380V	1st Landing Elevation	6.5m	
Frequency	50Hz	2nd Landing Elevation	15.615m	Access to the Portal Beam,
Rated Speed	0.7m/S	3rd Landing Elevation	47.845m	Access to the Operator's Cabin
Motor Power	11Kw	4th Landing Elevation	53.43m	Access to the Machinery House



Figure 1-59: Elevator



Figure 1-60: Elevator Car

1.8 Handling system

1.8.1 Spreader

As a "hand" of the crane, the spreader is the executive part of the crane to handle the containers. The spreader is designed to lift two 20' containers separated to 1.6 meter, with 50 tons capacity in single 40' mode and 65 (32.5x2) tons in twin operation.

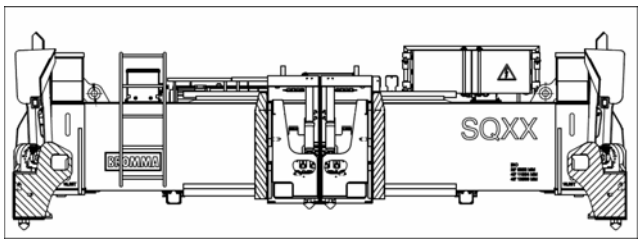


Figure 1-61: Spreader

1.8.2 Headblock

Four hoist sheaves mounted on the each corner of the headblock frame are the links between the hoist rope reeving system and the spreader and permanently reeved into the crane. It is a structural part of the crane and designed for specified crane capacity. The head block is furnished with pins for connecting to the spreader or the hook beam. The engagement or disengagement of the pins is monitored by proximity switches.



Figure 1-62: Headblock

1.8.3 Over-height adapter

A self-latching over-height adapter is provided. The over height attachment is designed to handle open top ISO containers, flat racks having over height loads and pallets with ISO fittings and dimensions. All functions are controlled by the master spreader.

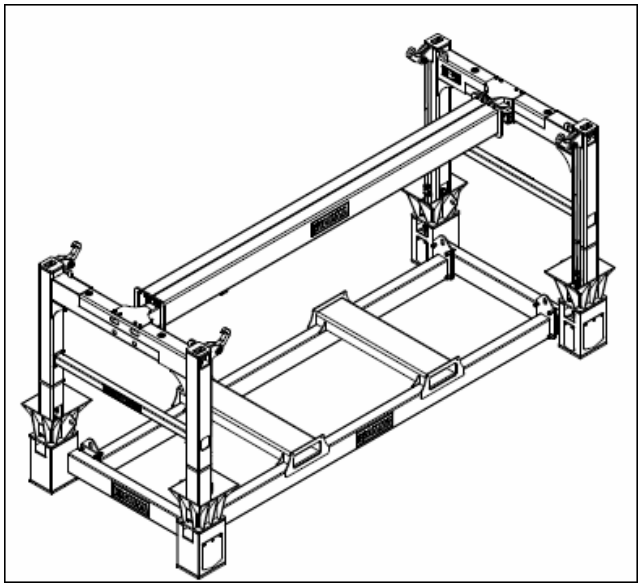


Figure 1-63: Over-Height AapterFrame

1.8.4 Hook beam

The hook beam of 75 tons rated capacity is designed to handle non-containerized loads. It is provided with a swivel safety single hook in the center of the beam. Attaching the sling to the hook and/or the pad eyes must be done by professionals.

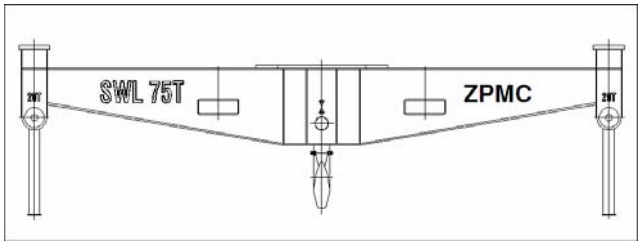


Figure 1-64: Hook Beam